

Using MOOCs to Reinvigorate Software Engineering Education (Keynote)

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ABSTRACT

The spectacular failure of the Affordable Care Act website ("Obamacare") has focused public attention on software engineering. Yet experienced practitioners mostly sighed and shrugged, because the historical record shows that only 10 % of large (>\$10 M) software projects using conventional methodologies such as Waterfall are successful. In contrast, Amazon and others successfully build comparably large and complex sites with hundreds of integrated subsystems by using modern agile methods and service-oriented architecture.

This contrast is one reason Industry has complained that academia ignores vital software topics, leaving students unprepared upon graduation. In too many courses, well-meaning instructors teach traditional approaches to software development that are neither supported by tools that students can readily use, nor appropriate for projects whose scope matches a college course. Students respond by continuing to build software more or less the way they always have, which is boring for students, frustrating for instructors, and disappointing for industry.

This talk explains how the confluence of cloud computing, software as a service (SaaS), and Massive Open Online Courses (MOOCs) have not only revolutionized the future of software, but changed it in a way that makes it *easier* and *more rewarding* to teach. UC Berkeley's revised Software Engineering course allows students to both enhance a legacy application and to develop a new app that matches requirements of non-technical customers, using the same tools and techniques that professionals use. By experiencing the whole software lifecycle repeatedly within a single college course, students learn to use and appreciate the skills that industry has long encouraged. The course is now popular with students, rewarding for faculty, and praised by industry.

The course is also the basis of a MOOC that over 100,000 students have taken. Through an arrangement with EdX, these materials are now available to classroom instructors wishing to teach this material, including lecture and review videos, automatic graders for assignments, and an inexpensive e-textbook (US\$10 / Rs400 / €8) that is matched section-for-section with the online materials. We and other

instructors have found that customizing the MOOC materials for classroom use as a SPOC (Small Private Online Course) is a promising model for curriculum technology transfer in the 21st century. The SPOC can greatly simplify instructor adoption, improve on-campus pedagogy, increase student throughput while raising course quality, and even reinvigorate faculty teaching. The speaker will offer a tutorial at ICSE 2014 on how instructors can use this material in their classrooms, backed up by an extensive instructor-support network including biweekly conference calls and an active community of instructors and content creators.

BIOGRAPHY

Armando Fox (fox@cs.berkeley.edu) is a Professor in Berkeley's Electrical Engineering & Computer Science Department as well as the Faculty Advisor to the UC Berkeley MOOCLab. He co-designed and co-taught Berkeley's first Massive Open Online Course on Engineering Software as a Service, currently offered through edX, through which over 10,000 students worldwide have earned certificates of mastery. He also serves on edX's Technical Advisory Committee, helping to set the technical direction of their open MOOC platform. His research is in improving student engagement and learning outcomes in MOOCs and on highly productive parallel programming (in the Berkeley ASPIRE Lab). While at Stanford he received teaching and mentoring awards from the Associated Students of Stanford University, the Society of Women Engineers, and Tau Beta Pi Engineering Honor Society. He has been a "Scientific American Top 50" researcher, an NSF CAREER award recipient, a keynote speaker at the Richard Tapia Celebration of Diversity in Computing, and an ACM Distinguished Scientist. In previous lives he helped design the Intel Pentium Pro microprocessor and founded a successful startup to commercialize his UC Berkeley Ph.D. research on mobile computing. He received his other degrees in electrical engineering and computer science from MIT and the University of Illinois.



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